

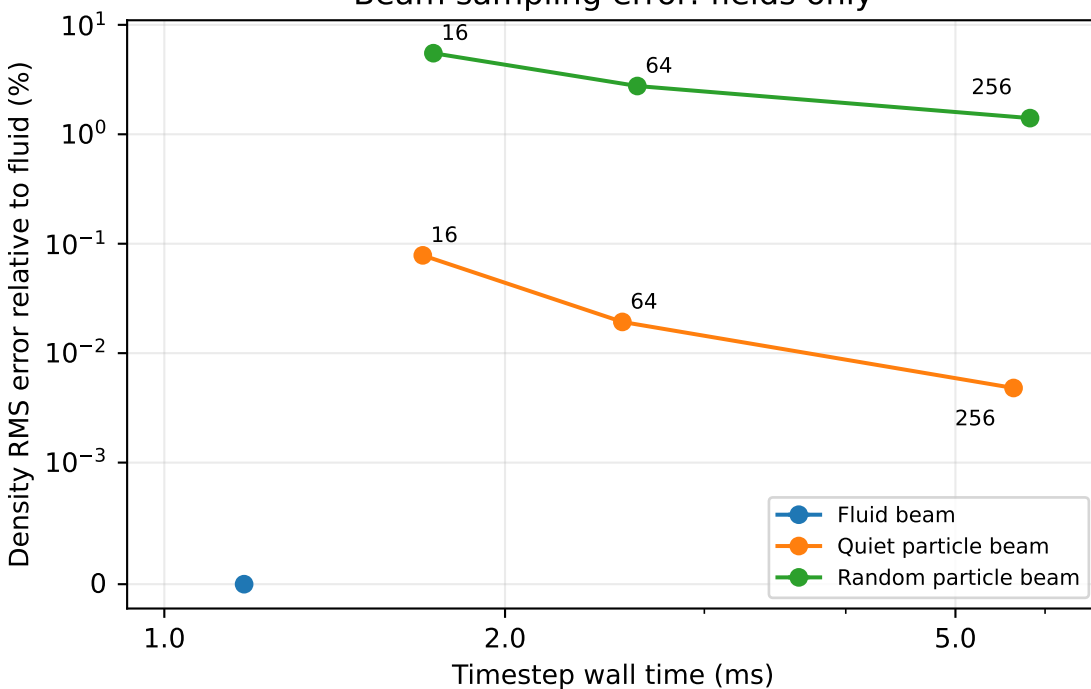
Perlmutter A100, RZ: measured noise and cost

Top: 64×256 , 6 seeds; left: 80 GB, right: 40 GB

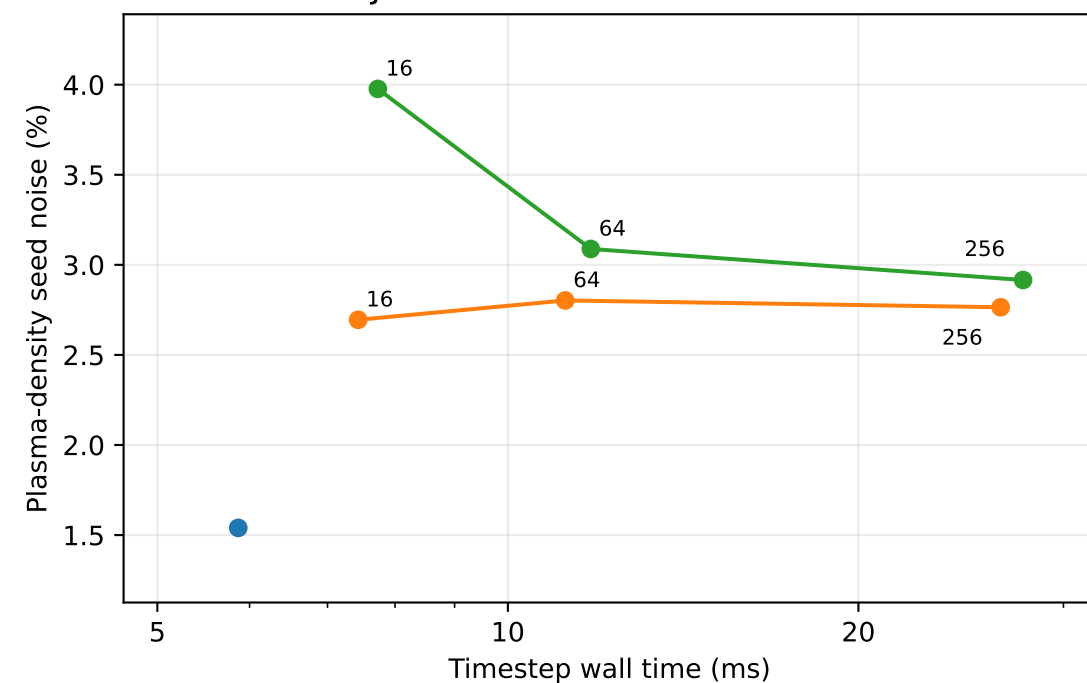
Bottom: 256×1024 , 3 seeds, 80 GB; quiet beam 64 particles/cell

Bars: 99% timing intervals; labels: beam particles/cell; synthetic MCC regression rates

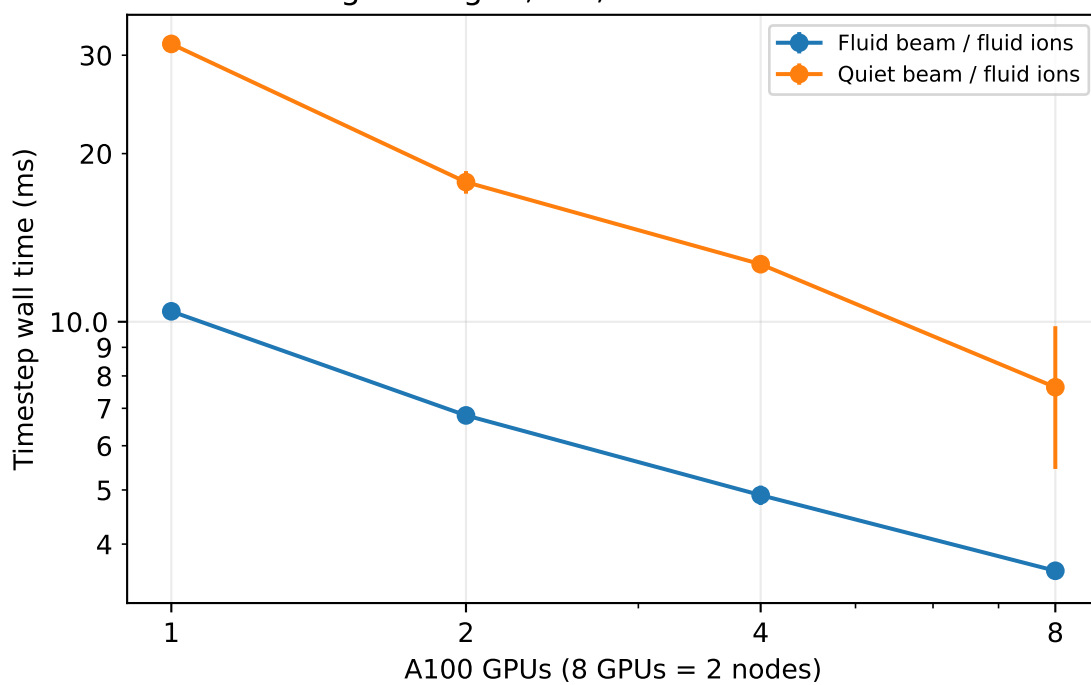
Beam sampling error: fields only



PJG + all electron MCC channels



Strong scaling: 1,048,576 identical electrons



Strong scaling: coupled production and MCC

